



NATIONAL TECHNICAL UNIVERSITY OF ATHENS
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
DIVISION OF COMPUTER SCIENCE

Continuous Machine Learning for Cooperative, Connected and Automated Mobility applications

DIPLOMA THESIS

of

GEORGIOS A. KYRIAKOPOULOS

Supervisor: Panayiotis Tsanakas
Professor

Athens, October 2025



NATIONAL TECHNICAL UNIVERSITY OF ATHENS
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
DIVISION OF COMPUTER SCIENCE

Continuous Machine Learning for Cooperative, Connected and Automated Mobility applications

DIPLOMA THESIS

of

GEORGIOS A. KYRIAKOPOULOS

Supervisor: Panayiotis Tsanakas
Professor

Approved by the three-member scientific committee on the 30th of October 2025

.....
Panayiotis Tsanakas
Professor NTUA

.....
Andreas-Georgios Stafylopatis
Professor Emeritus NTUA

.....
Georgios Alexandridis
Assistant Professor NKUA

Athens, October 2025

.....

Georgios A. Kyriakopoulos

Graduate of School of Electrical and Computer Engineering,
National Technical University of Athens

Copyright © **Georgios A. Kyriakopoulos, 2025.**

All rights reserved.

Copying, storage, and distribution of this work, in whole or in part, for commercial purposes is prohibited. Reproduction, storage, and distribution for non-profit, educational, or research purposes is permitted, provided that the source is cited and this notice is retained. Inquiries regarding the use of this work for commercial purposes should be directed to the author.

The views and conclusions expressed in this thesis represent those of the author and do not necessarily reflect the official positions of the National Technical University of Athens.

Περίληψη

Λέξεις Κλειδιά:

Abstract

Keywords:

Acknowledgements

First and foremost, I would like to express my sincere gratitude to my supervisor, **Professor Panayiotis Tsanakas**, for giving me the opportunity to conduct my thesis in a field that has been of great personal interest to me.

I would also like to thank **Dr. Georgios Drainakis** and **Dr. Panagiotis Pantazopoulos** of the **I-SENSE Research Group of ICCS/NTUA** for their valuable feedback and guidance towards the completion of this work.

I am grateful to my family and friends for their support and encouragement throughout this important chapter of my life, especially **Valentina** and **Artemis**, as well as **Nick**, **Apostolis**, **Alex**, **Christini**, and **Elina**.

Finally, and most importantly, I would like to thank **Serafeim** and **George**, with whom I shared the vision, the challenges, and the excitement of this work. Their collaboration and commitment were essential in shaping this thesis, and I truly appreciate sharing this journey with them.

Contents

Contents	11
List of Figures	13
List of Tables	15
Extended Abstract in Greek	17
1 Introduction	19
1.1 Problem Statement	19
1.2 Motivation	19
1.3 Thesis Goal	20
1.4 Contributions	20
1.5 Thesis Structure	21
2 Background	23
2.1 Classical Machine Learning	23
2.2 Deep Learning	24
2.3 Model Performance Metrics	24
2.4 Regularization, Overfitting, and Underfitting	25
2.5 Cross-Validation	25
2.6 Model Selection and Hyperparameter Tuning	26
2.7 Continuous Learning	26
2.8 MLOps	26
2.9 Concept Drift	27
2.10 Key Trade-offs in Drift Detection and Adaptation	28
2.11 SUMO Simulator	28
2.12 Containerization and Orchestration	29
2.13 Web Protocols and API Design	29
2.14 Backend and Frontend Frameworks	29
2.15 Data Validation and Storage	30
3 Relevant Work	31
3.1 Tooling Landscape for Drift Detection and Monitoring	31
3.1.1 River	31
3.1.2 scikit-multiflow	31
3.1.3 Alibi and Alibi Detect	32

3.1.4	Evidently	32
3.1.5	NannyML	33
3.2	ETA Prediction: Literature Overview	33
3.3	Novelty and Contributions of the Proposed Platform	35
Bibliography		41

List of Figures

List of Tables

Extended Abstract in Greek

Chapter 1

Introduction

1.1 Problem Statement

Urban mobility environments are highly dynamic and *non-stationary*, meaning that the statistical properties of traffic data change over time. Machine learning models trained on static historical data often assume a fixed data distribution, an assumption that rarely holds in real city conditions [1]. When deployed in a dynamic urban setting, a model’s performance can gradually degrade as driving patterns shift due to *concept drift*, the evolution of the underlying data relationships over time [2]. This drift is triggered by many factors: for example, road network changes (e.g., construction) can alter traffic flow patterns significantly [3]. Even routine variations like weather and seasonal demand swings can lead to abrupt or gradual shifts in vehicle speeds and congestion levels. A model that performs well on dry, low-traffic days may become less accurate during heavy rain or rush-hour peaks if those conditions were not part of its training distribution [4]. The challenge is that static ML models struggle to maintain accuracy in the face of non-stationarity, as their predictive power deteriorates over time unless the models continually adapt to the evolving urban mobility data [1]. Recognizing and addressing this concept drift in cooperative and automated mobility scenarios is therefore a critical problem identified by the research community.

1.2 Motivation

Given the inevitability of concept drift in real-world intelligent transportation systems, there is a clear need for a practical, drift-aware platform tailored to the mobility domain. While concept drift detection and adaptation have been studied in research, existing solutions are often fragmented or confined to laboratory settings [2, 5]. Current tools lack integration of critical components, for instance, traffic simulation, continuous monitoring, and adaptive retraining are rarely all incorporated into one seamless loop. This means practitioners may detect drifts, but they often have no automated way to replay scenarios, e.g. via simulation, or update models on the fly in response to those drifts [6, 7].

A domain-specific platform would be able to fill this gap by monitoring model performance through time and not just offline metrics, automatically detecting distribution shifts, and triggering model adaptation with minimal human intervention. Such an integrated approach is especially vital in *Cooperative, Connected and Automated Mobility (CCAM)* applications where

conditions can change rapidly and safety is paramount [8]. One key use case is *Estimated Time of Arrival (ETA)* prediction for vehicles, a core service in smart mobility that influences traffic management, logistics, and traveler information [9]. However, many current mobility systems do not continuously recalibrate ETA models as new data, e.g., live traffic feeds or weather updates, arrive [10]. This motivates the creation of a continuous learning platform for mobility: one that integrates traffic simulation to test “what-if” scenarios and generate synthetic data, live performance dashboards to observe errors as they evolve over time, drift detection algorithms to raise alerts when the model performance starts degrading, and an automated retraining pipeline to deploy updated models when needed. By addressing these needs, we aim to ensure that mobility prediction models remain accurate and robust despite the ever-changing urban environment.

1.3 Thesis Goal

The objective of this thesis is to design, implement, and evaluate a microservice-based platform for *continuous machine learning* in urban mobility applications. In essence, the goal is to create a *drift-aware ML platform* that supports end-to-end monitoring and adaptation of predictive models in a city traffic context.

The platform will be built using a modular microservice architecture to ensure scalability and flexibility, where each component (orchestration, model serving, drift detection, etc.) operates as an independent service. Crucially, the platform is intended to be model-agnostic and domain-specific: it can host any machine learning model for mobility prediction tasks, but this thesis will demonstrate and evaluate it primarily through the lens of an Estimated Time of Arrival (ETA) prediction use case. We focus on ETA because of its importance in CCAM scenarios, helping vehicles and travelers coordinate effectively, and because it provides a tangible benchmark to test how well the platform handles concept drift. Upon detecting drift, it will automatically trigger model adaptation, i.e. retraining and model swapping, to self-correct the predictions.

Although ETA is the main case study, the system is designed to accommodate additional models. For example, external predictive modules for Fuel Consumption [11] and Number of Stops [12] will be integrated to showcase that multiple concurrent mobility models can coexist and benefit from the shared drift-aware infrastructure. By the end of this thesis, we aim to show that such a platform can be realized and that it effectively maintains model performance over time in a realistic urban traffic setting.

1.4 Contributions

This thesis makes several contributions to the state of the art in continuous machine learning for mobility:

- **Synthetic Dataset Pipeline:** We developed a reproducible, modular data pipeline that generates training and evaluation datasets using synthetic traffic data from Simulation of Urban MObility (SUMO) [13]. Focusing on a case study of central Athens, the pipeline takes an open-source city road network and creates realistic traffic scenarios (vehicles, routes, events) to produce rich labeled data. This approach ensures that experiments are repeatable and tunable, meaning that the pipeline can be re-run for any city region or traffic pattern by adjusting parameters, thus addressing the shortage of standard drift benchmarks in urban mobility.

- **ETA Prediction Model under Drift:** We designed a feature-engineered ETA prediction model using gradient-boosted decision trees (LightGBM) [14]. The model is trained on the above dataset to estimate travel times for vehicles, and we introduce domain-specific features to improve accuracy. More importantly, the model’s performance is evaluated under simulated drift scenarios, e.g., a sudden shift from normal traffic to heavy rain conditions that reduce vehicle speeds. This allowed us to study how concept drift affects ETA prediction accuracy, and to validate that our model, and platform, can detect and respond to these changes.
- **Time-Lapse Drift Simulation:** We conducted a time-lapse simulation experiment to rigorously assess drift detection and adaptation in an end-to-end fashion. In this setup, the SUMO simulator replays an extended sequence where an initial period of stable conditions is followed by a clear drift event, in this case a transition from dry weather to a storm causing slower traffic. The platform’s drift detectors monitor the real-time prediction error and successfully detect the concept drift when the environment changes. We then measure how the system’s automatic adaptation, triggering model retraining with new data from the rainy period, helps recover the model’s performance. This experiment provides a realistic validation of the platform’s ability to observe, detect, and mitigate drift in a controlled yet lifelike scenario.
- **Drift-Aware Platform Implementation:** We present the design and implementation of a platform for continuous learning in mobility, which is a key practical contribution of this work. The platform includes components for simulation replay that feed historical or synthetic data through the model as if in real time, live dashboarding of model performance metrics, so that users can visualize concept drift as it happens, pluggable drift detection algorithms with calibration tools to set detection thresholds and minimize false alarms, and an automatic retraining and deployment pipeline, which seamlessly replaces the old model with an updated one when drift is confirmed. The entire system is built with modern software frameworks, such as containerized microservices, REST APIs for model serving, and event-driven triggers for retraining, to ensure it can be deployed in real operational environments. By open-sourcing the platform and detailing its architecture, we enable other researchers and practitioners to reproduce our approach and extend it to their own cooperative and automated mobility applications.

1.5 Thesis Structure

The remainder of this thesis is structured as follows:

- **Chapter 2: Background** - Introduces the foundational concepts and tools underlying our work. We review machine learning basics relevant to continuous model updating, the notion of concept drift, such as types of drift and common detection methods, the SUMO traffic simulator and its functionalities for creating mobility scenarios, and the platform technologies, like microservices, Docker, and APIs, used to build our solution.
- **Chapter 3: Related Work** - Surveys existing literature and systems in areas pertinent to our thesis. We cover drift detection tools and frameworks, examples being streaming ML libraries and MLOps solutions for model monitoring, examine available urban mobility simulation environments used to study drift, and review prior research on ETA prediction models in intelligent transportation. Finally, we discuss how our approach compares to and

innovates upon the current state of the art.

- **Chapter 4: Dataset and Modeling** - Details our methodology for creating the dataset and developing the ETA prediction model. We describe the data generation process using SUMO for the Athens case study, including how traffic demand patterns and drift scenarios, in this case weather changes, are configured. We then present the ETA model development, including feature engineering, hyperparameter tuning, model selection, and the overall training procedure and the set up of the experiments.
- **Chapter 5: Platform Architecture** - Provides a comprehensive overview of the drift-aware platform's design. We enumerate the microservices and their roles, explain how the system ingests data, either from simulation or future extensions to real streams, how drift detection algorithms are integrated, and how the platform orchestrates retraining and deployment of models. Implementation details of key components, such as the real-time dashboard, are given to illustrate how the system works in practice.
- **Chapter 6: Results** - Presents the results of our evaluation. We report the baseline performance of the ETA model under static conditions and then analyze its behavior under induced drift. We demonstrate the effectiveness of drift detection and quantify the gains from automatic adaptation. The results validate the platform's ability to maintain model accuracy over time.
- **Chapter 7: Conclusions** - Summarizes the contributions and findings of the thesis. We reflect on the success of the continuous learning approach for CCAM applications and any limitations observed. Finally, we outline future research directions, such as improvements to platform scaling and monitoring, or integrating the system with real-time data streams.

Chapter 2

Background

This chapter establishes the theoretical and technological foundation for the thesis. It begins by introducing essential concepts in machine learning, including classical and deep learning models, continuous learning paradigms, and model evaluation practices. It addresses operational concerns through MLOps principles, details the detection and adaptation to concept drift in dynamic environments, and explores the SUMO traffic simulator’s key functionalities and outputs used in this work. The chapter concludes with a review of modern software engineering technologies—such as Docker, FastAPI, Dash, Parquet, and RESTful APIs—that support reproducible experiments, scalable deployments, and robust data management within the broader system architecture

2.1 Classical Machine Learning

Machine learning (ML) is a field of artificial intelligence devoted to building algorithms that can learn from data patterns, enabling systems to predict and make decisions with minimal human intervention. ML algorithms are typically divided into four broad categories: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning [15].

- **Supervised learning:** involves training models on labeled datasets by learning a function that maps input features to output labels.
- **Unsupervised learning:** finds patterns and structures in unlabeled data, as seen in clustering or dimensionality reduction algorithms.
- **Semi-supervised learning:** combines small amounts of labeled data with larger unlabeled datasets, enhancing efficiency in domains with limited annotation.
- **Reinforcement learning:** guides an agent to optimal behavior through trial and error, rewarding desirable actions.

A few representative machine learning models are [16]:

- **Linear Regression:** Used for predicting continuous variables and modeling the relationship between features and output. It retains mathematical simplicity while remaining a crucial benchmark in both research and forecasting.
- **Decision Trees:** Represent decisions with branching structures, offering interpretability and handling mixed-type features.

- **Random Forests:** Ensembles of decision trees, reducing variance and improving generalization.
- **Support Vector Machines (SVM):** Classifies data by finding the optimal separating hyperplane, being effective in high-dimensional settings.
- **K-Nearest Neighbors (KNN):** Classifies points based on the “nearest” labeled examples.
- **Gradient Boosting Frameworks (XGBoost, LightGBM, CatBoost):** Ensemble methods that extend predictive power, speed, and robustness by aggregating the outputs of numerous decision trees to make predictions. XGBoost is renowned for its scalability and performance in structured/tabular data [17]. LightGBM excels at speed and memory efficiency [14]. CatBoost handles categorical variables and is competitive in ranking and classification tasks [18].

2.2 Deep Learning

Deep learning is a subset of machine learning that uses artificial neural networks to model and solve complex problems, inspired by the structure and function of biological neurons [19]. These networks are composed of multiple layers of neurons, each layer processing the output of the previous layer to extract increasingly complex features. A typical neural network comprises an input layer, one or more hidden layers, and an output layer.

A few representative deep learning models are:

- **Feedforward Neural Networks (Multi-Layer Perceptrons):** The core architecture for classifying or regressing fixed-size inputs.
- **Convolutional Neural Networks (CNNs):** Specialized for spatial relationships, such as image data.
- **Recurrent Neural Networks (RNNs):** Designed for sequential data, such as text or time series.
- **Transformer Models:** Modern architectures excelling in tasks ranging from natural language understanding to vision.

Neural networks are flexible and scalable but can be data-hungry, prone to overfitting, and require careful regularization and tuning to achieve optimal results.

2.3 Model Performance Metrics

Model evaluation is essential for assessing predictive reliability. Key metrics for regression models include [20]:

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

This metric conveys the average absolute prediction error, often valued for its interpretability and resistance to outliers.

- **Mean Absolute Percentage Error (MAPE):**

$$MAPE = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i|}$$

MAPE expresses error as a percentage relative to true values, facilitating comparisons across diverse scales. However, MAPE can be unstable when true values approach zero. Other metrics (RMSE, R^2) complement these in providing insight into the variance, fit, and reliability of regression and classification models.

- **Root Mean Squared Error (RMSE):**

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE measures the average magnitude of the prediction errors (residuals) between predicted and actual values in regression tasks. It is more sensitive to large errors than MAE due to squaring, penalizing outliers heavily.

- **R^2 (coefficient of determination):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

R^2 measures the proportion of variance in the target variable explained by the model, ranging from 0 to 1. Values near 1 indicate strong fit, while $R^2 \leq 0$ means the model performs worse than a constant baseline.

2.4 Regularization, Overfitting, and Underfitting

Models risk overfitting—memorizing the training set at the cost of poor generalization—when they become excessively complex. Underfitting happens when the model is too simple, missing essential patterns.

Regularization combats overfitting by penalizing complexity:

- **L1 Regularization (Lasso):** Encourages sparsity, zeroing non-critical coefficients.
- **L2 Regularization (Ridge):** Shrinks coefficients smoothly, preferred for stability.

In neural networks, additional techniques such as dropout, early stopping, and weight decay further mitigate overfitting.

2.5 Cross-Validation

Cross-validation is the gold standard for robust model evaluation. By partitioning data into training and validation folds, cross-validation (often k-fold) averages performance over multiple splits, reducing variance and protecting against accidental overfitting to a single dataset segmentation. Model selection and hyperparameter tuning typically leverage cross-validation to ensure real-world generalizability.

Some common cross-validation techniques are:

- **K-Fold Cross-Validation:** Partition the data into k equal folds, and then train on k-1 folds and validate on the remaining fold.

- **Stratified K-Fold Cross-Validation:** Partition the data into k equal folds, and then train on $k-1$ folds and validate on the remaining fold. The folds are stratified, meaning that the distribution of the target variable is kept the same in each fold.
- **Leave-One-Out Cross-Validation:** Partition the data into n equal folds, and then train on $n-1$ folds and validate on the remaining fold.
- **Repeated K-Fold Cross-Validation:** Repeat K-Fold Cross-Validation n times, and then average the performance over the n repetitions.

2.6 Model Selection and Hyperparameter Tuning

Selecting the right model and its hyperparameters is a critical decision in the ML workflow.

- **Model selection** often involves comparing candidate models (such as linear regression, decision tree, or boosting algorithms) using performance metrics evaluated on a validation set or through cross-validation.
- **Hyperparameter tuning** refers to optimizing parameters that govern model behavior but are not learned during training (e.g., learning rate, tree depth, regularization strength, number of layers in a neural network). Grid search systematically explores combinations, while random search samples the setup space. Advanced approaches such as Bayesian optimization build probabilistic models to efficiently home in on good hyperparameter settings.

2.7 Continuous Learning

Continuous learning, also known as continual or lifelong learning, refers to a paradigm in machine learning where models incrementally update their knowledge by learning from new data streams over time, rather than being trained solely once on a static dataset. In contrast to traditional retrain-from-scratch approaches, continuous learning enables models to adapt to evolving environments, integrate recent information, and remain relevant as data changes. Key strategies include incremental learning (updating parameters gradually), transfer learning, and experience replay.

A major challenge for continuous learning is preventing “catastrophic forgetting”, the loss of knowledge about previously learned tasks when adapting to new data [21]. Effective approaches balance learning new patterns (flexibility) with retaining established skills (stability).

2.8 MLOps

MLOps, short for Machine Learning Operations, is a discipline at the intersection of machine learning and DevOps, focused on standardizing, automating, and streamlining the end-to-end lifecycle of machine learning models in production environments. Its goal is to bridge the gap between data science and IT operations, helping organizations move ML prototypes from research to production efficiently and reliably.

Core MLOps principles include [22]:

- **Versioning:** Tracking code, data, models, and experiment metadata to ensure reproducibility and traceability across iterations and deployments.

- **Automation:** Implementing CI/CD (continuous integration/continuous delivery) pipelines for data ingestion, model training, validation, deployment, and rollback, reducing manual workload and errors.
- **Monitoring:** Continuously assessing live models for data drift, concept drift, and performance degradation, enabling timely retraining, adaptation, and governance.
- **Collaboration:** Facilitating teamwork between data scientists, engineers, and domain experts with standardized workflows, shared environments, and modular architectures.
- **Governance and Security:** Enforcing access controls, audit trails, and compliance with regulatory standards, especially in sensitive domains.

MLOps empowers organizations to deploy ML solutions at scale, react to switching data environments, and maintain high service quality and reliability. Challenges involve cultural change, tool integration, and evolving infrastructure requirements. Prominent MLOps platforms include cloud-native solutions (AWS SageMaker, Google Vertex AI), as well as open-source toolkits like MLflow and Kubeflow.

2.9 Concept Drift

Concept drift is the phenomenon where the statistical properties of the target variable change over time, leading to reduced model accuracy [2]. This effect is especially critical in streaming and non-stationary environments, such as online monitoring, finance, or cybersecurity.

Concept drift manifests in various forms:

- **Sudden (abrupt):** rapid changes in data distribution;
- **Gradual:** slow transitions between concepts;
- **Recurring (recurrent):** cyclic or seasonal patterns;
- **Incremental:** stepwise, small alterations over time.

Detection strategies are categorized as error-rate monitoring (e.g., DDM, EDDM, HLFR), statistical tests, and ensembles. Drift detectors can monitor prediction error, data distribution, or model performance metrics. Notable techniques include Hierarchical Hypothesis Testing (HHT), Linear Four Rates (LFR), and ensemble detectors, each with distinct strengths relevant to the type and speed of drift.

Accurate and timely drift detection is paramount for maintaining predictive accuracy and reliability. Systems often employ sliding windows or reference distributions to test for shifts, and performance degradation often triggers retraining and adaptation processes.

Concept drift adaptation strategies help machine learning models stay accurate as data changes over time. The main approaches include frequent retraining (updating the model on new data), online learning (incremental updates with each new sample), window-based methods (using only recent data for training), and ensemble methods (combining several models and favoring those performing best on recent data). Some strategies rely on drift detectors to trigger adaptation, while others update models continuously regardless of detected drift. The goal is to respond quickly to changes, maintaining reliable predictions in evolving environments.

2.10 Key Trade-offs in Drift Detection and Adaptation

Key trade-offs in drift detection and adaptation include [23]:

- **Sensitivity vs. False Alarms:** More sensitive detectors can spot small or early changes but risk frequent false positives. Less sensitive thresholds may delay detection, potentially degrading predictive performance.
- **Label Availability:** Some detectors require true labels to estimate error rates, limiting their use in unsupervised or delayed-label scenarios. Unsupervised methods can work on input distributions alone but often have less precision.
- **Latency:** Maintaining low-latency detection is crucial for real-time systems but may require sacrificing some detection accuracy or threshold tightness to avoid delays.
- **Computational Cost:** More advanced or frequent drift checks, especially those recalculating statistics or performing window comparisons, can significantly raise resource requirements, impacting scalability in high-volume streams.

Carefully balancing these trade-offs is essential when choosing drift detectors and adaptation strategies for different operational contexts, such as real-time monitoring, batch analytics, or resource-constrained environments.

2.11 SUMO Simulator

SUMO (Simulation of Urban MObility) is an open-source, microscopic and continuous multi-modal traffic simulation suite designed to model and analyze the movement of individual vehicles, pedestrians, and various transportation modes on large-scale road networks [24]. Each agent, be it a car, bus, bicycle, or pedestrian is simulated explicitly, with unique routes, behaviors, and interactions, allowing for fine-grained analyses of traffic dynamics and management strategies [25].

Core functionalities and capabilities include:

- **Microscopic Simulation:** Models each vehicle and pedestrian individually, supporting realistic behaviors such as lane changes, right-of-way rules, and traffic light interactions.
- **Multi-modal Traffic:** Simulates not only private cars but also public transport, bicycles, pedestrians, and intermodal trips within the same environment.
- **Flexible Network Import:** Supports importing road networks from OpenStreetMap, commercial formats (VISUM, Vissim, NavTeq), and others, or allows users to create synthetic networks.
- **Demand Generation:** Generates vehicle flows using origin-destination matrices, real traffic counts, or synthetic (random and virtual population-based) demand models.
- **Traffic Management:** Enables the testing and evaluation of traffic lights, routing policies, eco-aware navigation, and urban policies before real-world deployment.
- **Real-time Control and Interoperability:** Offers APIs like TraCI for live, programmatic control of the simulation and integration with external tools, such as vehicle communication (C2X) simulators.

- **Comprehensive Visualization:** Provides tools for route finding, visualization and detailed performance monitoring.
- **Extensive Simulation Outputs:** Generates vehicle trajectories, trip statistics, traffic states, environmental metrics, and more in XML or CSV, enabling detailed analysis.
- **High Performance and Scalability:** Efficiently manages large networks (10,000+ edges; 100,000+ vehicles) and supports parallel simulations on various platforms (Windows, Linux, macOS).
- **Open Source and Extensible:** Freely available and extensible, with a large user and developer community for enhancements, custom models, and research applications.

These features make SUMO an established tool for transportation planning, intelligent transportation system development, traffic light optimization, and research on smart mobility solutions worldwide.

2.12 Containerization and Orchestration

Docker is a widely adopted containerization platform that allows the packaging of software and its dependencies into isolated units called containers [26]. This isolation ensures that code runs the same way regardless of the underlying environment, solving the notorious “it works on my machine” problem. Docker containers are lightweight, highly portable, and enable versioned environments. Academic use cases include reproducible computational research, easy sharing of code and data, and simplifying complex setups for fellow researchers. In large organizations and regulated industries (e.g. healthcare, finance), Docker provides essential security, agility, and compliance by isolating sensitive data and quickly deploying new services.

Docker Compose builds on Docker by orchestrating multiple containerized services (such as databases, backends, frontends) through simple configuration files [27]. This facilitates local development of microservice architectures, comprehensive integration testing, and staging realistic production environments before deployment.

2.13 Web Protocols and API Design

HTTP is the foundational protocol for web and API communications, enabling clients and servers to exchange requests and responses in a standardized format, including JSON and XML payloads.

REST APIs are the core architectural style in web services for interoperable communication over HTTP [28]. By strictly adhering to stateless client-server principles, REST APIs allow disparate systems, ML pipelines, data stores, external user interfaces, to work together seamlessly.

2.14 Backend and Frontend Frameworks

FastAPI is a modern web framework for building RESTful APIs with Python [29]. Distinctively, FastAPI leverages Python type hints and asynchronous programming, allowing developers to write concise, robust, and extremely fast endpoints. Its native compatibility with OpenAPI and automatic documentation greatly reduces manual effort and errors. FastAPI is widely used in both prototypes and mission-critical systems due to its speed and strong validation.

Dash is a Python framework for creating interactive web dashboards [30]. With minimal Python code, data scientists and engineers can build sophisticated analytics apps that integrate visualizations, user inputs, and RESTful API endpoints. Dash leverages Flask under the hood, but developers can extend with additional Python frameworks as required for interoperable, customized solutions.

Plotly is an extensible visualization library available on top of Dash, useful for interactive charts, data exploration, and embedding analytics in dashboards [31].

2.15 Data Validation and Storage

Pydantic is the leading Python library for data parsing and validation using type hints [32]. By providing strict enforcement of schemas at runtime, Pydantic allows developers to safely handle incoming data—whether from users, APIs, or databases—preventing subtle bugs and security risks. Models inherit from `BaseModel` and use Python’s type annotations to declaratively describe fields, their types, and validation logic. When instantiating a model, Pydantic parses and validates input, raising structured errors for invalid data. Its integration with FastAPI allows for automatic generation of API documentation and validation of request and response data.

Apache Parquet is a column-oriented data format optimized for analytical workloads in big data settings [33]. Parquet’s design enables highly efficient compression and encoding, dramatically reducing storage costs and query times, especially compared to row-based formats like CSV.

Chapter 3

Relevant Work

3.1 Tooling Landscape for Drift Detection and Monitoring

3.1.1 River

River is an open-source Python library for online machine learning on streaming data, born from the merger of the Creme and scikit-multiflow projects. It provides a single unified framework to train models incrementally, one sample at a time, which makes it well-suited to scenarios with continuous data and potential concept drift. River includes a variety of learning algorithms (e.g. linear models, Hoeffding trees, ensemble methods) that can update their parameters on the fly, as well as evaluators and metrics that update incrementally along with the model. Notably, River treats concept drift as a first-class concern: it implements plug-and-play drift detection methods like ADWIN, DDM, and EDDM to signal when the statistical properties of the data or the model’s error rate have changed significantly. A major strength of River is its efficiency and low overhead for streaming applications, it avoids expensive retraining by doing continuous learning, enabling rapid adaptation to changing data. However, River’s focus on single-instance updates means it may not directly leverage batch acceleration or GPU computing as efficiently as some batch frameworks. Moreover, building deep neural models in River is less common, so for very complex patterns (e.g. image or graph data) one might need external solutions. Overall, River is focused on streaming adaptation and concept drift handling, making it a powerful tool for real-time ML systems[34].

3.1.2 scikit-multiflow

scikit-multiflow is an earlier Python framework for stream learning and concept drift, which was one of River’s predecessors (Montiel et al., 2018)[35]. Inspired by the MOA stream mining software in Java, scikit-multiflow brought popular stream learning algorithms and evaluation tools into the Python ecosystem. It supports a range of tasks including classification, regression, and even multi-output prediction, and it provides several drift detection algorithms (e.g. ADWIN, DDM, EDDM, Page-Hinkley) as part of its toolkit. A key design goal was compatibility with scikit-learn, allowing users to integrate stream learning with familiar APIs. The strengths of scikit-multiflow include its breadth of implemented methods (many inherited from the MOA framework) and the ability to simulate data streams with built-in generators for controlled concept drift (e.g. abrupt or gradual drift in synthetic data). One limitation, however, is that scikit-multiflow is no longer as actively maintained; its functionality has effectively been subsumed into River, which

offers a more consolidated and improved interface. In practice, new projects prefer River for streaming tasks, but scikit-multiflow remains a reference point in the literature and is still useful for benchmarking new drift detection techniques due to its comprehensive implementation of classical methods.

3.1.3 Alibi and Alibi Detect

Alibi is an open-source library by Seldon Dev focused on machine learning model inspection and explainability, offering techniques like counterfactual explanations and influence scores for black-box models (e.g. explaining why a prediction was made)[[AlibiExplainability](#)]. Building on this, the Seldon team also released Alibi Detect, which is a library specifically for outlier detection and concept drift detection in ML applications [github.com](#). Alibi Detect provides a collection of both offline and online detectors for various data types: tabular data, text, images, and time series [github.com](#). It includes statistical methods (like Kolmogorov-Smirnov or Chi-square tests for distribution drift in features) as well as learned detectors (like drift detection using neural network embeddings for complex data such as images) [github.com](#). A notable focus of Alibi Detect is support for different modalities and integration with modern ML stacks: it can use TensorFlow or PyTorch under the hood, which allows it to implement advanced detectors (e.g. a PCA-based outlier detector or a classifier-based drift detector) that leverage GPUs. The strength of Alibi/Alibi Detect lies in its broad coverage: it tackles not just drift but also outliers and even adversarial example detection in one package, making it a flexible choice for production monitoring where explainability and data quality are needed alongside drift detection [github.com](#). One limitation is that Alibi Detect is a lower-level library – it provides algorithms but not a full monitoring solution; users must integrate it into their workflow (for example, scheduling periodic drift checks or responding to detector alerts in an MLOps pipeline). Additionally, configuring the more complex detectors may require expert knowledge (for instance, setting thresholds or choosing embedding models). In summary, Alibi (for explainability) and Alibi Detect (for drift/outliers) offer a robust toolkit focused on model insight and data drift, well-suited for use cases where understanding model behavior and detecting distribution shifts are both important (Looveren et al., 2020)[36].

3.1.4 Evidently

Evidently is an open-source Python library and toolset designed to evaluate, test, and monitor machine learning models and data in production [medium.com](#). It provides pre-built reports and dashboards to analyze things like data drift, target drift, data quality, and model performance over time [medium.com](#) [medium.com](#). One of Evidently’s core features is the ability to generate an interactive HTML dashboard comparing a reference dataset (for example, the training data or last week’s data) to a current dataset; the dashboard includes visualizations and statistical tests that highlight any significant changes in feature distributions or model outputs. Under the hood, Evidently employs statistical drift detection metrics suitable for each feature type (e.g., Kolmogorov–Smirnov test for continuous features, Jensen-Shannon divergence or chi-square for categorical features) [evidentlyai.com](#) [medium.com](#). It also tracks model performance metrics (if ground truth is available) or proxy metrics (like prediction score distributions) as part of its monitoring reports. The strengths of Evidently include its ease of use and rich visualization – with minimal code, a user can produce a comprehensive report that would otherwise require manual analysis. This makes it a popular choice for integrating into Jupyter notebooks or automated pipelines to regularly check for data drift and model decay. Evidently also supports customizable

test suites: users can define assertions (e.g., “no feature drift beyond a certain threshold”) and get alerts when these are violated [medium.com](#) . A current limitation is that Evidently’s real-time capabilities are evolving; it was initially built for batch analysis and reporting. Using it in a truly live monitoring scenario (continuous streaming data) may require additional effort (such as writing a job to compute metrics on rolling windows). Additionally, while it identifies drift and performance issues, it doesn’t automatically retrain models or suggest fixes – it’s primarily a monitoring dashboard. In short, Evidently fills the role of ML monitoring and data validation by making drift detection and model quality checks more accessible and transparent (Evidently AI, 2023)[37].

3.1.5 NannyML

NannyML is a newer open-source library specifically focused on post-deployment model monitoring, with an emphasis on detecting concept drift and estimating model performance without immediate ground truth (Grootendorst et al., 2023)[**NannyML**]. It aims to answer the question: “How is my model performing now, given that I might not have labels for the latest predictions?” To this end, NannyML provides tools for data drift detection (monitoring changes in input feature distributions) as well as algorithms to infer performance drop. A centerpiece is its Confidence-Based Performance Estimation (CBPE) method, which uses the model’s prediction probabilities and detected data drifts to estimate metrics like accuracy or ROC AUC in production before real labels arrive [dev.to](#) . In practical terms, NannyML can alert you that “your model’s inputs have shifted significantly, and it estimates your model’s accuracy has dropped from (say) 0.85 to 0.70” – allowing proactive retraining or investigation [dev.to](#) . Another innovative feature is the “reverse drift” (RCD) concept: assessing how changes in input data would be expected to affect the target distribution and thus the model’s error, essentially linking drift to business impact [nannyml.com docs.nannyml.com](#) . The strength of NannyML lies in this performance-centric approach to drift: it goes beyond flagging that data has changed, by quantifying the likely effect on model predictions and outcomes. This is particularly useful in industries where obtaining true labels is slow or expensive (e.g., credit risk, where you only know defaults after many months). NannyML’s limitations include that it currently supports primarily classification tasks (binary and multiclass) for its performance estimation module [nannyml.readthedocs.io](#) . The techniques can be complex to calibrate – for instance, CBPE assumes the model’s probability estimates are well-calibrated and that past relationships hold, which might not always perfectly predict actual performance. Additionally, as a specialized library, it may need to be integrated into an existing MLOps setup (it provides Python APIs and some UI components, but not a full monitoring server on its own). In summary, NannyML represents an advanced approach to concept drift monitoring: instead of just detecting drift, it focuses on what that drift means for model accuracy, helping close the gap between drift signals and model maintenance actions [dev.to](#) .

3.2 ETA Prediction: Literature Overview

Predicting travel time or estimated time of arrival (ETA) is a well-studied problem in the transportation domain, and a variety of machine learning approaches have been explored. Early works approached ETA prediction with classical time-series models like ARIMA, treating travel time as a time-dependent process; while such statistical models worked in limited settings, they struggled with the inherent non-linearities and context dependencies of traffic data [frontiersin.org](#) . Over the past decade, the field has seen a surge of deep learning methods that capture complex spatial and temporal patterns in traffic networks. For example, recurrent neural

networks (RNNs) and long short-term memory (LSTM) architectures have been used to model vehicle trajectories as sequences of GPS points, successfully learning temporal dependencies such as road segment travel times and rush-hour effects (Duan et al., 2016; Shen et al., 2019)[38, 39]. Convolutional neural networks have been applied to encode spatial information — one notable approach is to treat a path’s GPS sequence like an “image” or matrix (e.g., via a grid or along-route segmentation) and use CNNs to extract features (Wang et al., 2018)[40]. More recently, attention mechanisms and transformers have been introduced: AttentionTTE by Li et al. (2024) employs self-attention to capture global interactions among road segments, combined with a recurrent module for local temporal trends [frontiersin.org](#) [frontiersin.org](#) . This achieved state-of-the-art results on a large trajectory dataset, indicating the benefit of modeling both local and long-range dependencies in routes [frontiersin.org](#) [frontiersin.org](#) . Similarly, researchers have explored multi-task learning in deep models – for instance, the WDR (wide-deep-recurrent) model and CoDriver system incorporated an auxiliary task (learning driver behavior style) to improve ETA predictions, demonstrating that additional contextual signals can improve accuracy [frontiersin.org](#) .

In parallel to sequence-based methods, graph-based approaches have gained prominence for ETA and traffic prediction. In a road network, intersections and road segments can be naturally represented as nodes and edges in a graph, so graph neural networks (GNNs) can explicitly model the connectivity and traffic flow between road segments. Derrow-Pinion, Xu, Layne, and al. [41] describe a GNN-based ETA model deployed in production for Google Maps, which learns from massive amounts of traffic data while incorporating road network topology and dynamic conditions (like accidents or rush hour patterns). By using GNN layers to propagate information along the road graph, their model can account for upstream slowdowns or congestion that classical models might miss [arxiv.org](#) . This graph-based ETA estimator yielded significant improvements in real-world accuracy – for example, reducing large ETA errors by over 40% [arxiv.org](#) . Other studies have followed similar ideas, building spatio-temporal graph models that combine GNNs with temporal sequence modeling (e.g. graph convolution plus LSTM) to predict travel times under varying conditions (Jia et al., 2021; Guo et al., 2022)[42, 43]. These advanced deep learning models represent the current state-of-the-art, especially when rich data is available (historical trajectories, real-time traffic sensors, etc.). They tend to excel at capturing complex patterns, but at the cost of higher complexity and data requirements.

Despite the success of deep neural approaches, tabular data methods remain highly relevant, especially in industry settings where interpretability, simplicity, or limited data are considerations. Many practical ETA prediction systems reduce the problem to a regression on engineered features: for example, features might include the route distance, expected traffic speed, number of traffic lights, time of day, day of week, weather conditions, and so on. Gradient boosting decision trees (GBDT) have been a popular choice for such tabular formulations (Guin, 2006; Zhang and Haghani, 2015)[44, 45]. GBDT models (implemented in tools like XGBoost or LightGBM) offer strong performance on structured data and have the advantage of producing feature importance insights, which is valuable for understanding ETA drivers. Zhang and Haghani (2015) demonstrated that with carefully constructed input features (including real-time traffic indices), a GBDT model can achieve accuracy comparable to more complex models, though it may need frequent updating to handle drift [frontiersin.org](#) . In recent work, Antypas et al. (2022)[46] applied XGBoost, CatBoost, and LightGBM to predict bus arrival times in a smart city context, showing that these boosted tree models are competitive for ETA and can be tuned to high accuracy given domain-specific feature engineering. In their case, gradient boosting achieved around 30% [researchgate.net](#) . The continued popularity of gradient boosting in this domain is due

to several factors: (i) Tabular features can encode a lot of prior knowledge (for instance, the road speed limit or historical average travel time on a segment is an informative predictor that a tree can easily use); (ii) training and inference are fast and resource-efficient compared to deep sequence models, which is useful for deployment on edge devices or with limited infrastructure; and (iii) the models are easier to maintain (one can retrain a new boosted tree model on recent data periodically, and examine feature importances if the model starts to degrade). Of course, a limitation is that these models do not automatically capture sequential or spatial relationships unless those are manually turned into features. Therefore, the current trend in ETA research and applications is often a hybrid one: use domain knowledge to engineer strong features and/or combine outputs of simpler models, and, where possible, incorporate the structure of the problem (sequential or graph) via specialized models or ensemble approaches. Well-known benchmarks, such as the NYC Taxi trip duration dataset (Kaggle 2015) or the Google Maps ETA data (used in CIKM 2021 Challenge), often show that gradient boosting with rich features can be very effective, coming close to deep learning methods when the latter are not heavily optimized. In summary, ETA prediction methods span from deep learning models leveraging spatio-temporal structure to gradient-boosted tree models on tabular features, and the choice often depends on available data and the need for interpretability versus maximum accuracy.

3.3 Novelty and Contributions of the Proposed Platform

The platform developed in this thesis brings together ideas from concept drift research, MLOps, and domain-specific traffic simulation to create a closed-loop adaptive learning system for mobility. Here we summarize the key novel contributions of this platform, and how it differentiates itself from existing tools and literature:

1. **Controlled Drift Simulation with SUMO:** In order to study concept drift in a realistic yet controlled manner, we integrate a traffic simulation (SUMO – Simulation of Urban MObility) into the platform. SUMO allows us to generate synthetic traffic data under varying conditions with fine control. A novel aspect is the creation of a time-lapse scenario in which we deliberately introduce a sudden change in environment: for example, the platform simulates 10 hours of normal driving conditions followed by 10 hours of heavy rain, where rain is modeled by globally reducing the road friction coefficient in SUMO. This causes vehicles to slow down and drive differently, thereby inducing concept drift in the input data and the relationship between features (e.g. speeds, distances) and targets (travel time, etc.). By having this capability to manufacture drift on demand, we can evaluate how well different detectors and adaptation strategies perform on a known “ground truth” drift event. Traditional drift detection research often relies on either real-world events (which are unpredictable) or synthetic data generators for drift (e.g., rotating hyperplane datasets) jmlr.csail.mit.edu. Our use of SUMO bridges that gap by producing realistic, domain-specific drift: it’s not just abstract changes in data distribution, but changes grounded in a physical scenario (weather change in traffic) that a transportation agency would care about. This approach is relatively unique – to our knowledge, few if any published works have demonstrated a full ML workflow where a traffic microsimulator is used to inject controlled drifts and then automatically detect and counteract them in real time. Moreover, the platform continuously tracks the model’s performance during the simulation; as the rain scenario unfolds, we log the model’s prediction error metrics (e.g. mean absolute error for ETA) to observe how performance degrades due to drift. This provides an authentic demonstration of concept drift’s impact on an ML model deployed in a dynamic environment.

2. Multi-Task Prediction and Monitoring: The platform is designed to handle multiple related prediction tasks simultaneously, which is a departure from most drift detection case studies that focus on a single prediction output. In our implementation, a single model (or a set of coordinated models) produces predictions for ETA, fuel consumption, and number of stops for a given vehicle trip. This multi-task setup reflects a more complex operational scenario (for instance, a smart mobility service might want to predict not just arrival time but also fuel usage and stops for logistics planning). The contribution here is twofold: (a) from a modeling perspective, the system can either use multi-output regression models or separate models that share features – either way, it must manage and monitor several predictions at once; and (b) from a drift detection perspective, the platform monitors drift on multiple data streams and multiple performance metrics concurrently. For example, during the rain-induced drift, ETA predictions might degrade first while fuel predictions remain stable, or vice versa, so our system watches each task. This is novel compared to existing tools like River or NannyML which typically assume a single target. Our platform demonstrates how a drift in the environment can have different manifestations across tasks (e.g., rain might drastically affect ETA and fuel consumption but not the predicted number of stops if stops depend more on route layout than speed). It required developing a custom monitoring strategy that can aggregate signals from multiple drift detectors and multiple outputs. We implemented a simple consensus mechanism: the drift detection micro-service in our system runs detectors on input features (data drift) and on each model’s error stream (performance drift), and only if a consistent drift signal is observed (for instance, error rate for ETA spiking and a known change in input distribution) do we declare a drift event and trigger adaptation. This reduces false alarms in a noisy multi-task setting. Supporting multiple tasks in one platform is a practical contribution because real-world systems often have to monitor dozens of metrics – our work provides a template for how to achieve that in an automated way.

3. Automated Model Retraining and Hot-Swapping: Perhaps the most significant contribution of the platform is the seamless closed-loop adaptation capability. When concept drift is detected (either via a statistical test or a drop in performance beyond a threshold), the platform doesn’t just raise an alert – it initiates a retraining pipeline and deployment of an updated model, without human intervention. Concretely, the system maintains a buffer of recent data (e.g., the last 1 hour of vehicle trips) and when drift is confirmed, it triggers a retraining job that fine-tunes or re-fits the ML model using this new data distribution. We implement this efficiently for the gradient boosting model used for ETA: since training from scratch can be done quickly on the updated dataset (which is not huge in our case), we opted for a simple re-training rather than online learning updates, to ensure we capture the new pattern (e.g., slower speeds in rain) comprehensively. Once the new model is trained and validated on a hold-out set, the platform performs a hot swap: the running system begins using the new model for all subsequent predictions, with minimal downtime. Techniques from continuous deployment are used here – for example, the old model and new model might briefly run in parallel to ensure the new one is warmed up, and then a switch is flipped (this can be done via a REST API call to the predictor service to load the new model). This kind of automated retraining and deployment is often discussed in MLOps but not commonly demonstrated end-to-end in academic literature [researchgate.net](https://www.researchgate.net) . Recent work by Abhay et al. (2025)[47] proposed an automated pipeline using Evidently to detect drift and trigger retraining, deploying the new model if it outperforms the old – our platform realizes a similar idea, but in the specific context of mobility and streaming data, and combines it with the aforementioned drift simulation. The novelty here is showing the entire loop: we not only detect drift, we close the loop by adapting to it immediately. This required solving engineering challenges (e.g., ensuring that the system can retrain while continuing to serve

predictions, avoiding a halt) and design challenges (deciding when retraining is worth it to avoid model churn). The outcome is an architecture where model performance degradation is not only observed but automatically countered by an update – moving towards the ideal of a self-healing ML system.

4. Real-Time Dashboard and Visualization: To make the system’s operations interpretable and to aid in human-in-the-loop oversight, the platform includes a real-time dashboard that visualizes the current state of the simulation and the ML models. This dashboard (built with Dash/Plotly) displays key information such as: current traffic conditions in the simulation (e.g., a map of Athens with vehicles, possibly highlighting rain conditions), the live predictions vs. ground truth for ETA and other tasks, and drift detection metrics over time (for instance, a timeline showing the p-value of a drift test or the model’s error rate). When a drift is detected and a model swap occurs, the dashboard visibly flags this event (e.g., “Drift detected at 10:30, model retrained and updated at 10:35”), and subsequent improvements in error can be observed. While a dashboard itself is not a novel research contribution, integrating it end-to-end with the platform is a valuable demonstration for practitioners. It closes the feedback loop by allowing users to watch the ML system adapt in real time, thereby building trust. Many existing drift tools like Evidently produce static reports or require the user to inspect logs; in contrast, our system’s UI shows a continuously updating view, which is especially important in a streaming context. This also helps communicate the results: for example, stakeholders can see that during the rain scenario the original model’s error kept rising, but as soon as the retrained model was deployed, the error drops back to normal levels – effectively illustrating the benefit of the adaptive approach.

In summary, the platform distinguishes itself by combining drift generation, detection, and adaptation in one unified environment. Unlike existing monitoring libraries that stop at detection, or online learning frameworks that update models but often on synthetic data, our system covers the full lifecycle on a realistic traffic problem: it manufactures a drift (rainy weather) in a mobility digital twin, observes the model’s decline, catches it via multiple detectors, and then remedies it through automated retraining – all while delivering insights through a real-time interface. This end-to-end closed-loop approach, especially applied in the context of cooperative, connected, and automated mobility, is a novel contribution demonstrating how advanced MLOps techniques (continuous monitoring, automatic retraining) can be applied to maintain model performance in non-stationary environments. It effectively serves as a prototype for next-generation intelligent traffic management systems where AI models remain reliable over time by continually learning from new conditions. The lessons from this platform could be extended to other domains as well, but are particularly pertinent to mobility where change (in weather, human behavior, infrastructure) is constant and must be managed by adaptive learning systems.

Bibliography

- [1] Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., and Bouchachia, A. “A survey on concept drift adaptation”. In: *ACM Computing Surveys (CSUR)* 46.4 (2014), pp. 1–37.
- [2] Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., and Zhang, G. “Learning under concept drift: A review”. In: *IEEE Transactions on Knowledge and Data Engineering* 31.12 (2018), pp. 2346–2363.
- [3] Maheshwari, H., Yang, L., and Pazzi, R. W. “Machine Learning Advancements in Urban Traffic Simulation: A Comprehensive Survey”. In: *IEEE Open Journal of Intelligent Transportation Systems* 6 (2025), pp. 1027–1052.
- [4] Baier, L., Hofmann, M., Kühl, N., Mohr, M., and Satzger, G. “Handling Concept Drifts in Regression Problems—the Error Intersection Approach”. In: *arXiv preprint arXiv:2004.00438* (2020).
- [5] Greco, S., Vacchetti, B., Apiletti, D., and Cerquitelli, T. “Unsupervised Concept Drift Detection from Deep Learning Representations in Real-time”. In: *arXiv preprint arXiv:2406.17813* (2025).
- [6] Xin, H., Zhang, X., Tang, R., Yan, S., Zhao, Q., Yang, C., Cui, W., and Yang, Z. “LitSim: A Conflict-aware Policy for Long-term Interactive Traffic Simulation”. In: *arXiv preprint arXiv:2403.04299* (2024).
- [7] Modesto, C., Borges, J., Nahum, C., Matni, L., Both, C. B., Cardoso, K., Gonçalves, G., Correa, I., Lins, S., Silva, A., and Klautau, A. “Towards a Robust Transport Network With Self-adaptive Network Digital Twin”. In: *arXiv preprint arXiv:2507.20971* (2025).
- [8] ERTRAC Working Group “Connectivity and Automated Driving”. *Connected, Cooperative and Automated Mobility Roadmap*. Tech. rep. Version 10. European Road Transport Research Advisory Council (ERTRAC), 2022.
- [9] Abdi, A. and Amrit, C. “A Review of Travel and Arrival-Time Prediction Methods on Road Networks: Classification, Challenges and Opportunities”. In: *PeerJ Computer Science* 7 (2021), e689.
- [10] Arifi, A., Bouros, P., and Chondrogiannis, T. “A Study on ETA Prediction using Machine Learning and Recovered Routes”. In: *Proceedings of the Workshops of the EDBT/ICDT 2024 Joint Conference*. CEUR Workshop Proceedings. Paestum, Italy: CEUR-WS.org, 2024.
- [11] Angelis, G. *Fuel Consumption Prediction Model*. Unpublished work, integrated in drift-aware platform. 2025.
- [12] Tzelepis, S. *Number of Stops Prediction Model*. Unpublished work, integrated in drift-aware platform. 2025.
- [13] Krajzewicz, D., Erdmann, J., Behrisch, M., and Bieker, L. “Recent development and applications of SUMO - Simulation of Urban MObility”. In: *International Journal On Advances in Systems and Measurements* 5.3&4 (2012), pp. 128–138.

- [14] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T. “LightGBM: A Highly Efficient Gradient Boosting Decision Tree”. In: *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. 2017, pp. 3146–3154.
- [15] Shalev-Shwartz, S. and Ben-David, S. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.
- [16] Sarker, I. H. “Machine Learning: Algorithms, Real-World Applications and Research Directions”. In: *SN Computer Science* 2.160 (2021). DOI: 10.1007/s42979-021-00522-x.
- [17] Chen, T. and Guestrin, C. “XGBoost: A Scalable Tree Boosting System”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2016, pp. 785–794. DOI: 10.1145/2939672.2939785.
- [18] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. “CatBoost: unbiased boosting with categorical features”. In: *Advances in Neural Information Processing Systems 31 (NeurIPS)* (2018), pp. 6638–6648.
- [19] LeCun, Y., Bengio, Y., and Hinton, G. “Deep learning”. In: *Nature* 521 (2015), pp. 436–444. DOI: 10.1038/nature14539.
- [20] Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006.
- [21] Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., and Wermter, S. “Continual lifelong learning with neural networks: A review”. In: *Neural Networks* 113 (2019), pp. 54–71. DOI: 10.1016/j.neunet.2019.01.012.
- [22] Tabassam, A. I. U. “MLOps: A Step Forward to Enterprise Machine Learning”. In: *arXiv preprint arXiv:2305.19298* (2023). URL: <https://arxiv.org/abs/2305.19298>.
- [23] Zliobaitė, I. “A survey on concept drift adaptation”. In: *ACM Computing Surveys (CSUR)* 46.4 (2014). DOI: 10.1145/2523813.
- [24] Foundation, E. and contributors. *Eclipse SUMO - Simulation of Urban MObility*. <https://eclipse.dev/sumo/>. Accessed 2025-10-26. 2024.
- [25] Espinosa, A. V. “Traffic Modeling with SUMO: a Tutorial”. In: *arXiv preprint arXiv:2304.05982* (2021). URL: <https://arxiv.org/abs/2304.05982>.
- [26] Merkel, D. “Docker: lightweight Linux containers for consistent development and deployment”. In: *Linux Journal* 239 (2014), p. 2.
- [27] Inc., D. *Docker Compose Documentation*. <https://docs.docker.com/compose/>. Accessed 2025-10-26. 2025.
- [28] Fielding, R. T. “Architectural Styles and the Design of Network-based Software Architectures”. In: *Doctoral dissertation, University of California, Irvine* (2000). URL: https://www.ics.uci.edu/~fielding/pubs/dissertation/rest_arch_style.htm.
- [29] Ramírez, S. and contributors. *FastAPI Documentation*. <https://fastapi.tiangolo.com/>. Accessed 2025-10-26. 2025.
- [30] Inc., P. T. *Dash Documentation & User Guide*. <https://dash.plotly.com/>. Accessed 2025-10-26. 2025.
- [31] Inc., P. T. *Plotly Python Documentation*. <https://plotly.com/python/>. Accessed 2025-10-26. 2025.
- [32] Colvin, S. and contributors. *Pydantic Documentation*. <https://docs.pydantic.dev/>. Accessed 2025-10-26. 2025.
- [33] Vohra, D. “Apache Parquet”. In: *Practical Hadoop Ecosystem*. Apress, 2016, pp. 325–351.
- [34] Montiel, J., Halford, M., Mastelini, S. M., Bolmier, G., Sourty, R., Vaysse, R., Zouitine, A., Guillet, F., Read, J., Abdessalem, T., and Bifet, A. “River: Machine Learning for Streaming Data in Python”. In: *Journal of Machine Learning Research* 22.110 (2021), pp. 1–8. URL: <https://www.jmlr.org/papers/v22/20-1380.html>.

- [35] Montiel, J., Read, J., Bifet, A., Abdesslem, T., and Zliobaite, I. “Scikit-multiflow: A Multi-output Streaming Framework”. In: *Proceedings of the International Conference on Machine Learning (ICML) Workshop on Automated Machine Learning*. 2018. URL: <https://github.com/scikit-multiflow/scikit-multiflow>.
- [36] Looveren, A. V. and Klaise, J. *Alibi Detect: Algorithms for Outlier, Adversarial and Drift Detection*. <https://github.com/SeldonIO/alibi-detect>. Version 0.9.4. 2020.
- [37] AI, E. *Evidently: Evaluate, Test and Monitor Machine Learning Models*. <https://github.com/evidentlyai/evidently>. Version 0.3.2. 2023.
- [38] Duan, Y., Lv, Y., and Wang, F.-Y. “Travel Time Estimation Based on LSTM Neural Network”. In: *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*. 2016, pp. 153–158. DOI: 10.1109/ITSC.2016.7795560.
- [39] Shen, Y., Xiao, Z., Zeng, H., Ye, Y., and Xu, H. “DeepTransport: Learning Spatial-Temporal Dependency for Traffic Condition Forecasting”. In: *IEEE Transactions on Intelligent Transportation Systems* 21.11 (2019), pp. 4607–4617. DOI: 10.1109/TITS.2019.2904751.
- [40] Wang, Y., Chen, J., Zhu, Y., Zhang, Y., Zhang, C., Zheng, Y., and Yu, P. S. “When Will You Arrive? Estimating Travel Time Based on Deep Neural Networks”. In: *AAAI Conference on Artificial Intelligence*. Vol. 32. 1. 2018. URL: <https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/view/16076>.
- [41] Derrow-Pinion, A., Xu, F., Layne, C., and al., et. “ETA Prediction with Graph Neural Networks in Google Maps”. In: *CIKM ’21: Proceedings of the 30th ACM International Conference on Information and Knowledge Management*. 2021, pp. 3767–3776. DOI: 10.1145/3459637.3482246.
- [42] Jia, B., Cao, Y., Song, Z., and Wang, S. “A Spatio-Temporal Graph Convolutional Network for Travel Time Estimation”. In: *IEEE Access* 9 (2021), pp. 129403–129414. DOI: 10.1109/ACCESS.2021.3113696.
- [43] Guo, S., Lin, Y., and al., et. “Learning Dynamic Graph Structure for Traffic Forecasting”. In: *IEEE Transactions on Knowledge and Data Engineering* (2022). DOI: 10.1109/TKDE.2022.3165185.
- [44] Guin, A. *Travel Time Prediction Using Gradient Boosting Decision Trees*. Georgia Institute of Technology Master’s Thesis. 2006.
- [45] Zhang, L. and Haghani, A. “A hybrid model for real-time bus arrival time prediction”. In: *Transportation Research Part C: Emerging Technologies* 60 (2015), pp. 420–435. DOI: 10.1016/j.trc.2015.09.006.
- [46] Antypas, D., Papakostas, A., Georgoudakis, D., and Xydis, S. “Predicting Bus Arrival Time for Autonomous Shuttles in Smart Cities Using Tree-based Models”. In: *2022 International Conference on Smart Applications, Communications and Networking (SmartNets)*. 2022, pp. 1–8. DOI: 10.1109/SmartNets55927.2022.9966784.
- [47] Abhay, A. and Johnson, K. *Automatic Retraining Pipeline for Drift-Adaptation using Evidently and MLflow*. <https://evidentlyai.com/blog/drift-detection-auto-retraining-pipeline>. Under review. 2025.